

Day 1 Assignment — Dart Basics

Variables, types, string interpolation, and arithmetic

Before you start

- Everything here can be done at `dartpad.dev`. Nothing to install.
- Write the code yourself. Do not copy from ChatGPT — you will only be cheating your own practice.
- Run every program before you submit. If it does not run, it is not finished.
- Total: 50 marks. Expect this to take 45–60 minutes.

Question 1 — Your Profile Card

8 MARKS

Write a program that stores the following information in variables, each with the correct explicit type (do not use `var` here):

- Your full name
- Your age
- Your height in metres
- Whether you are currently a student (true or false)

Then print exactly two lines using string interpolation:

```
Name: <your name>, Age: <your age>
Height: <your height>m, Student: <true or false>
```

Question 2 — Fix the Errors

10 MARKS

Each snippet below has exactly one mistake. Write the corrected line and, in a comment, explain in one sentence what was wrong.

(a)

```
int price = 99.50;
```

(b)

```
string city = 'Kathmandu';
```

(c)

```
var count = 10;
count = 'ten';
```

(d)

```
int age = 24
print(age);
```

(e)

```
String name = 'Aarav';
int score = 88;
print('$name scored $score + 10 marks');
// This should print: Aarav scored 98 marks
```

Question 3 — Predict the Output

8 MARKS

Do not run these first. Write down what you think each one prints, then run it and note whether you were right. Marks are for the reasoning, not just the answer.

(a)

```
print(10 / 2);
```

(b)

```
print(17 ~/ 5);
print(17 % 5);
```

(c)

```
int a = 7;
int b = 2;
print('Result: ${a / b}');
```

(d)

```
var x = 5;
print('Value is $x + 1');
```

Question 4 — Rectangle Calculator

8 MARKS

Store a length of 12.5 and a width of 8 in variables with appropriate types. Calculate and print the area and the perimeter. Your output should look like this:

```
Length: 12.5, Width: 8
Area: 100.0
Perimeter: 41.0
```

All three lines must use string interpolation. Do the calculation inside the interpolation for at least one of them.

Question 5 — Shop Bill

10 MARKS

A customer buys the following at a shop:

Item	Quantity	Price per unit (Rs.)
Notebook	3	45
Pen	5	12
Bag	1	850

Write a program that stores all of this in variables and prints a bill:

```
--- BILL ---
Notebook x3 = Rs. 135
Pen x5 = Rs. 60
Bag x1 = Rs. 850
Subtotal: Rs. 1045
Tax (13%): Rs. 135.85
Total: Rs. 1180.85
```

Do not hard-code any of the calculated numbers. Every total must be computed from the variables.

Question 6 — Seconds to Time

6 MARKS

A video is 3725 seconds long. Store that in a variable and print how long it is in hours, minutes, and seconds:

```
3725 seconds = 1h 2m 5s
```

Hint: you will need ~/ and %. Think about what each one gives you.

Bonus — Your Own Question

+5 MARKS

Write one question of your own that uses everything from today, and solve it. It should be something a classmate could attempt. Interesting questions get read out in the next class.

How to submit

- One .dart file per question, named q1.dart , q2.dart , and so on.
- Put them in a folder called day-01-assignment .
- Share it in the group before the next class. From next week onward, submissions go to GitHub.
- Stuck? Try for 20 minutes first, then ask in the group. Post your code and the error, not just “it’s not working”.